Test Series No.

03

NEET Success Kit [2026-27]

DURATION: 195 Minutes | **M. MARKS: 720**

DATE: 10/06/2026

SYLLABUS

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Botany:	Full Syllabus
Zoology:	Full Syllabus

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 15 minutes** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

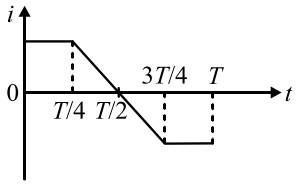
Name of the Student (In CAPITALS): _____

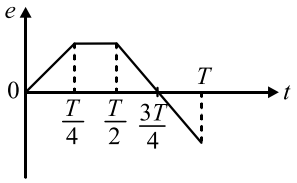
Roll Number: _____

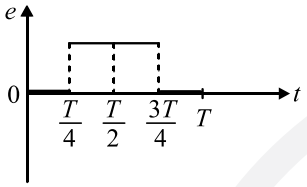
OMR Bar Code Number: _____

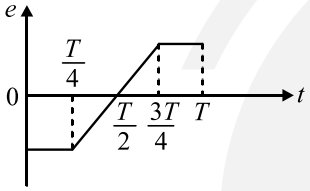
Student's Signature: _____ **Invigilator's Signature:** _____

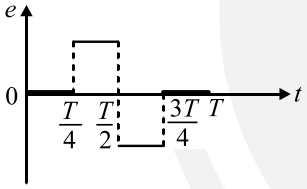
1. The following figure shows the variation of current i in a coil with time t. The corresponding variation of induced emf e with time t is:



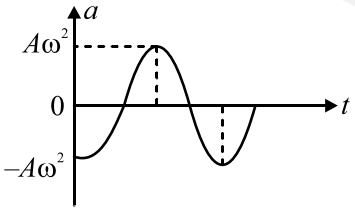
(1)


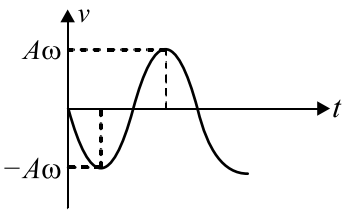
(2)


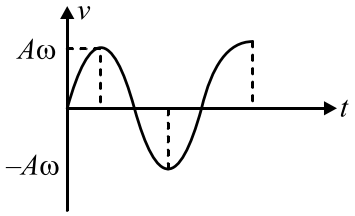
(3)


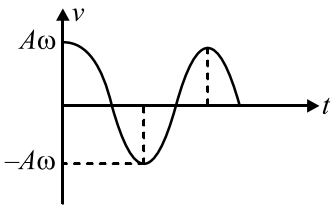
(4)


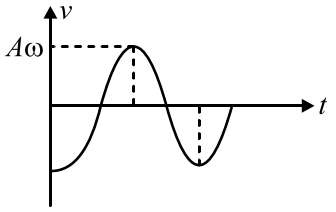
2. Acceleration-time (a-t) graph for a simple harmonic oscillator is shown below. Identify the corresponding velocity-time graph. (symbols have their usual meaning)



(1)


(2)


(3)


(4)


3. Increase in forward bias/applied voltage in a P-N junction diode results in:

(1) increase in the width of depletion region

(2) decrease in the width of depletion region

(3) decrease in the flow of majority charge carriers across the junction.

(4) increase in the flow of minority charge carriers across the junction.

4. A plane electromagnetic wave propagates in a vacuum along the positive z-direction. At a specific coordinate and time, the electric field vector is measured to be 12 V/m along the positive x-axis. What is the magnetic field vector at that point?

(1) 4×10^{-8} T along negative y-direction

(2) 4×10^{-8} T along positive y-direction

(3) 3.6×10^{-9} T along positive y-direction

(4) 3.6×10^{-9} T along negative y-direction

5. An infinite charged sheet is placed in a region of space. The ratio of magnitude of electric field due to the sheet at two points with respective perpendicular distances 10 cm and 30 cm from the sheet will be:

(1) 1 : 3 (2) 3 : 1

(3) 1 : 1 (4) 9 : 1

6. Arrange the speed of UV rays (v_{uv}) visible light (v) and IR rays (v_{IR}) in vacuum.

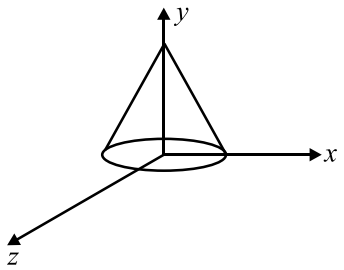
(1) $v_{uv} < v < v_{IR}$

(2) $v_{uv} > v > v_{IR}$

(3) $v_{uv} = v = v_{IR}$

(4) $v_{uv} = v_{IR} > v$

7. A cone of height 2 m and radius of base 1 m is kept in a region of space as shown in figure. The electric field is uniform and is given by $7\hat{j}$ N/C. The net inward electric flux-through the cone is:



(1) zero (2) -7π Vm

(3) $\frac{-14}{\pi}$ Vm (4) -2π Vm

8. The power factor for an ideal choke coil connected to an AC source is:

(1) zero

(2) 1

(3) same as power factor of a purely capacitive AC circuit

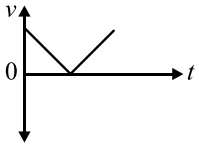
(4) Both (1) and (3)

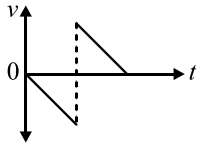
9. A boy stands on a weighing machine which is placed on a table and ties the machine with his feet. The reading on the machine shows 200 N. Now he attempts to jump from the table on ground with the machine tied. The reading on the machine while he is in air will be: (g = 10 m/s^2)

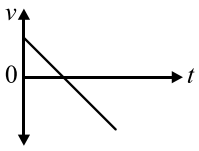
(1) 200 N (2) 20 N

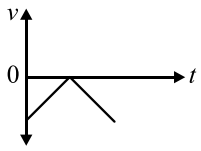
(3) 2000 N (4) zero

10. A body is dropped from a height in the absence of air resistance. It strikes the ground elastically and rebounds back upwards. The correct velocity (v) *vs* time (t) graph for its motion would be: (Take upward direction positive)

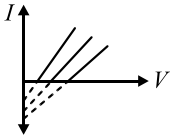
(1) 

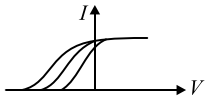
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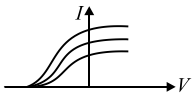
(3) 

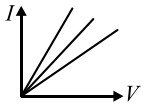
(4) 

11. Variation of photoelectric current (I) with applied potential difference (V) for three different frequencies of incident radiation but a fixed incident intensity is described by which of the following graph?

(1) 

(2) 

(3) 

(4) 

12. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Houses made of concrete roofs get very hot during summer days.

Reason R: Thermal conductivity of concrete is not small enough to keep the room cool.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false

(2) A is false but R is true

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

13. The escape velocity from the Earth's surface is v. The escape velocity from the surface of another planet having a radius five times that of Earth and same mass density is:

(1) $5v$ (2) $\sqrt{5}v$

(3) $\frac{v}{\sqrt{5}}$ (4) $25v$

14. If YDSE apparatus is immersed in a liquid of refractive index 1.6, then fringe width will:

(1) becomes 1.6 times

(2) becomes 0.625 times

(3) remains same

(4) becomes (1.6)^2 times

15. If the speed of an electron in 5th Bohr orbit of an hydrogen atom in v, then the speed of electron in 2nd orbit for Li^{++} is:

(1) $7.5v$ (2) $2.5v$

(3) $5v$ (4) $0.3v$

[Contd...

16. A highly elastic wire has a Poisson's ratio of 0.4. Upon applying a longitudinal stretching force, its cross-sectional area decreases uniformly by 2.4%. What is the corresponding percentage elongation of the wire?

(1) 1.5% (2) 3.0%

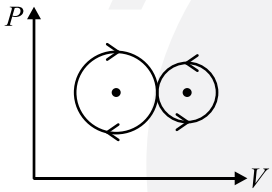
(3) 4.8% (4) 6.0%

17. Water flows through a horizontal pipe. At a point where the speed of water is v, the pressure is P. At another point where the speed is $2v$, the pressure will be (density of water = ρ):

(1) $P-\frac{3}{2}\rho v^2$ (2) $P-\rho v^2$

(3) $P-2\rho v^2$ (4) $P+\frac{3}{2}\rho v^2$

18. An ideal gas describes a process shown by the following indicator diagram. The net amount of work done by gas is:



(1) Positive (2) Negative

(3) Zero (4) Infinite

19. A massless rope is suspended from a rigid support with weight 12 N in figure *A*. It is then tied to another rope and pulled with a force of 5N such that it makes an angle θ with vertical as shown in figure *B*. The ratio of tension T_1 and T_2 in the two cases is:

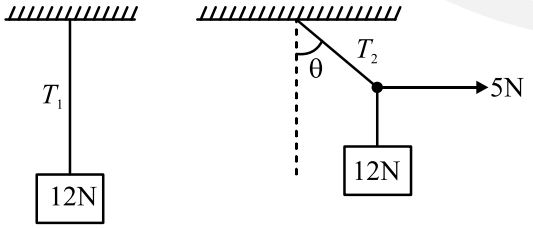


Fig. A Fig. B

(1) 12 : 13 (2) 13 : 12

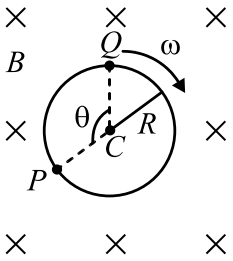
(3) 12 : 5 (4) 5 : 12

20. A straight wire of length 5 m and carrying current 1 A is suspended mid-air by a uniform magnetic field of magnitude 4.9 T acting perpendicular to the wire. The mass of wire is:

(1) 2 kg (2) 3 kg

(3) 3.5 kg (4) 2.5 kg

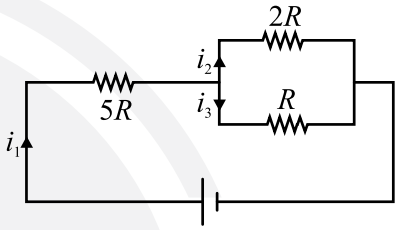
21. A metal disc of radius R is rotating with a constant angular velocity ω in a region of uniform magnetic field B as shown in figure. The induced emf across points P and Q is given by:



(1) $\frac{1}{2}B\omega R^2$ (2) $B\omega R^2\cos\theta$

(3) $B\omega R^2\sin\theta$ (4) zero

22. In the given circuit, the ratio of current $i_1 : i_2 : i_3$ is:



(1) 5 : 1 : 2 (2) 2 : 5 : 10

(3) 3 : 1 : 2 (4) 1 : 2 : 5

23. Identify the **INCORRECT** statement for interference.

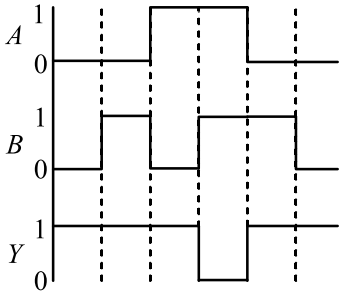
(1) In YDSE interference for coherent sources the intensity of 3rd bright fringe is same as that of 5th bright fringe.

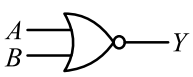
(2) In YDSE interference for coherent sources the width of central bright fringe is double that of other bright fringes.

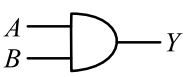
(3) sound wave also show interference

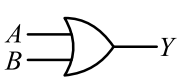
(4) During constructive interference of two sources with amplitude A each, resultant amplitude is $2A$.

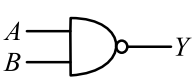
24. Identify the **correct** logic gate corresponding to the given input and output waveforms.



(1) 

(2) 

(3) 

(4) 

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25. Given below are two statements:

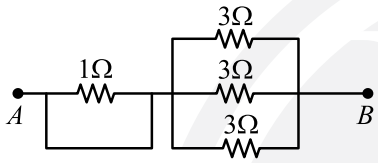
Statement I: When the plates of a parallel plate capacitor are drifted apart while connected to the battery, the potential difference across plates decreases.

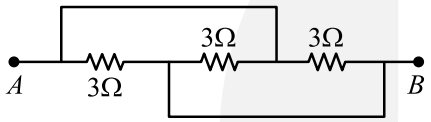
Statement II: The capacitance of a parallel plate capacitor is inversely proportional to the separation between the plates.

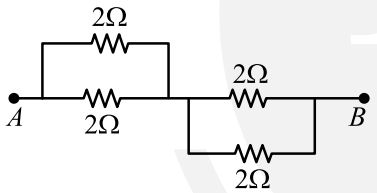
In the light of the above statements, choose the *most appropriate* answer from the options given below:

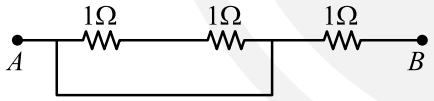
(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

26. In which of the following circuit(s), the effective resistance between A and B is equal to 1Ω.

I. 

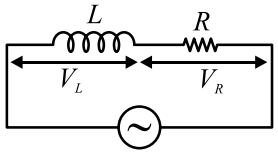
II. 

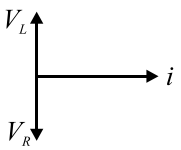
III. 

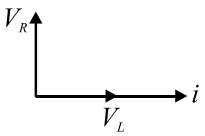
IV. 

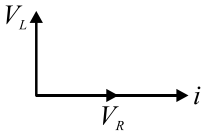
(1) I and II only
(2) I, II and IV only
(3) II and III only
(4) II and IV only

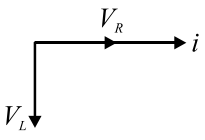
27. Identify the **correct** phasor diagram corresponding to the given *L.R.* AC circuit. (symbols have usual meaning)



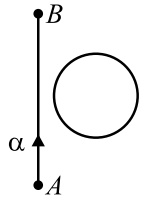
(1) 

(2) 

(3) 

(4) 

28. An alpha particle moves on a straight line path from A to B as shown in figure. What will be the direction of induced current in the coil kept adjacent to the path of alpha particle and in the same plane?



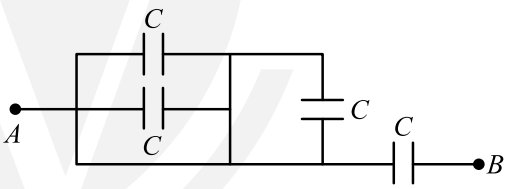
(1) first anticlockwise then clockwise
(2) first clockwise then anticlockwise
(3) clockwise throughout
(4) anticlockwise throughout

29. In a transverse progressive wave of amplitude A, the magnitude of wave velocity is half of the magnitude of maximum particle velocity. The wavelength of the wave is:

(1) πA (2) $\frac{\pi A}{2}$

(3) $2\pi A$ (4) $\frac{2A}{\pi}$

30. The equivalent capacitance between A and B of the combination of capacitors each having capacitance C shown in the figure is:



(1) $\frac{C}{2}$
(2) $2C$
(3) $\frac{3C}{4}$
(4) C

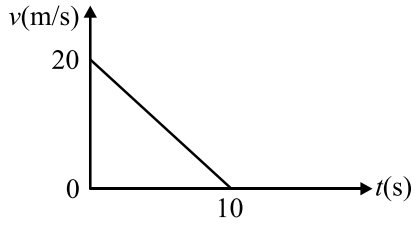
31. A bar magnet oscillates in a uniform magnetic field with a potential energy of U. It is cut perpendicular to its length into two equal parts. These two pieces are then stacked one over another with like poles together. The potential energy of the new system (in the same magnetic field and at the same orientation) is:

(1) $\frac{U}{2}$ (2) U

(3) $2U$ (4) $4U$

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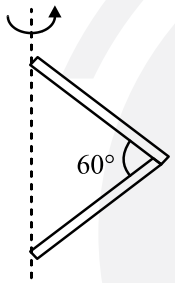
32. Velocity-time graph of a particle of mass 5 kg moving on a straight line is as shown in figure. Work done by all the forces on the particle in time interval $t = 0$ to $t = 10$ s is:



(1) 1000 J (2) – 1000 J

(3) 750 J (4) – 750 J

33. Two identical rods shaped to form a 'V' with an angle 60° between them are made to rotate about an axis passing through their free ends as shown in figure. If the mass of each rod is 2 kg & length is 1 m, the moment of inertia of the system is:



(1) 1 kg m^2

(2) 2 kg m^2

(3) 3 kg m^2

(4) 4 kg m^2

34. Light with an average flux of 10 W/cm^2 falls on a non-reflecting surface at normal incidence having surface area 50 cm^2. The energy received by the surface during time span of 2 minutes is:

(1) 4×10^4 J

(2) 6×10^4 J

(3) 5×10^4 J

(4) 2×10^4 J

35. A blackbody at temperature T radiates heat per unit area at the rate of R. If its temperature becomes $2T$ and surface area is doubled, the rate of heat radiated per unit area will be:

(1) $16R$

(2) $8R$

(3) $4R$

(4) $32R$

36. A ray of light is incident at an angle of 30° on one surface of a small angled prism and emerges normally from the opposite surface. If refractive index of material of prism is 1.5, then sine of angle of prism is:

(1) 0.33

(2) 0.15

(3) 0.5

(4) 20

37. A projectile is fired at an angle θ (< 90°) on a horizontal surface. If its trajectory is described by the equation $y = x - \frac{x^2}{40}$ (where x and y are in meter), the maximum height attained by the projectile is: (Take x-axis as horizontal & y-axis as vertically upwards)

(1) 10 m

(2) 15 m

(3) 20 m

(4) 40 m

38. A nucleus at rest disintegrates into two smaller nuclei with their masses in the ratio of 2:1. After disintegration they will move:

(1) In opposite directions with speed in the ratio of 1 : 2 respectively

(2) In opposite directions with speed in the ratio of 2 : 1 respectively

(3) In the same direction with same speed.

(4) In opposite directions with the same speed.

39. A screw gauge covers 0.2 cm on the linear scale while making two complete rotations of circular scale. If number of circular scale divisions 50, the least count of the screw gauge is:

(1) 0.2 mm

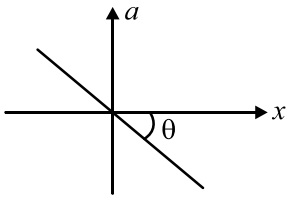
(2) 0.02 mm

(3) 0.1 mm

(4) 0.01 mm

[Contd...

40. Time period of a simple harmonic oscillator is 2π s. The value of θ in the given acceleration displacement graph for the oscillator will be:



(1) 30° (2) 60°
(3) 45° (4) 90°

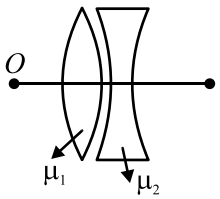
41. An elevator has a fully loaded mass of 1200 kg. It moves upward at a constant speed of $2.5\ m\,s^{-1}$. If a constant frictional drag of 3000 N opposes its motion, determine the minimum power the motor must deliver. (Assume $g = 10\ m\,s^{-2}$)

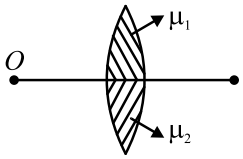
(1) 30.0 kW (2) 42.5 kW
(3) 45.0 kW (4) 37.5 kW

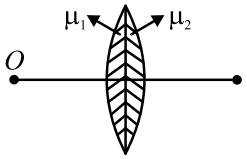
42. A ball of mass 200 gm is dropped from a height of 20 m straight on the horizontal surface of a bat which rebounds it to a height of 31.25 m. The magnitude of impulse imparted by the bat to the ball will be: (g = 10 m/s^2)

(1) 10 kg m/s (2) 9 kg m/s
(3) 5 kg m/s (4) 1 kg m/s

43. Which of the following arrangements of optical thin lenses will produce two images of an object placed in front at position O. (refractive indices $\mu_1 \neq \mu_2$)

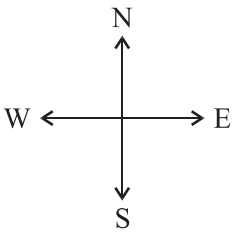
A.


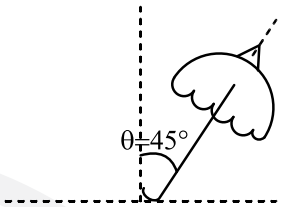
B.


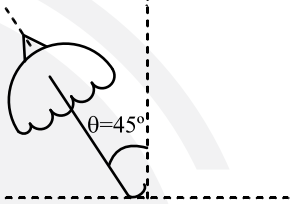
C.


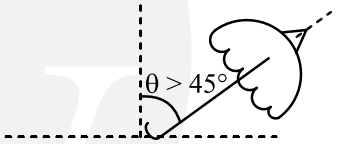
(1) B Only
(2) C Only
(3) A and C only
(4) A, B and C

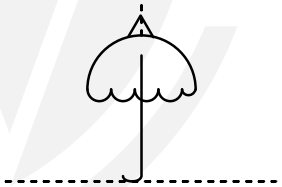
44. Rain is falling in the vertically downwards direction. Suddenly wind starts blowing towards west with speed same as rain. If a man is walking towards east in rain. What should be the possible direction of his umbrella to avoid maximum rain?



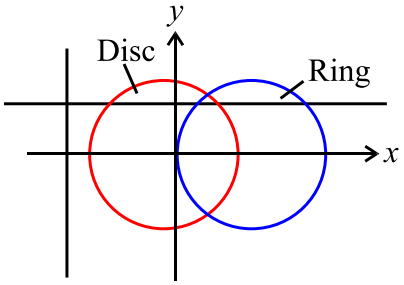
(1)


(2)


(3)


(4)


45. A uniform thin ring of mass M and radius R is placed on a uniform thin disc of mass $2M$ and radius R such that the ring passes through the center of the disc. The distance of the center of mass of the entire system from the center of the disc is:



(1) $\frac{R}{2}$ (2) $\frac{R}{3}$
(3) $\frac{2R}{3}$ (4) $\frac{R}{4}$

[Contd...]

46. Consider the following statements about the temperature dependence of rate constants:

I. The Arrhenius equation is valid only for elementary homogeneous reactions.

II. In the Arrhenius equation, the unit of A is the same as that of k.

III. At a given temperature, a lower E_a corresponds to a faster reaction.

IV. The parameters A and E_a used in the Arrhenius equation are temperature-dependent.

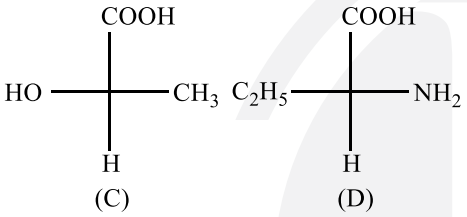
V. When $E_a >> RT$, A and E_a become interdependent.

Choose the **correct** set:

(1) II and III (2) I, III and IV

(3) II, IV and V (4) I and II

47. Determine the absolute configurations of (C) and (D).



(C) (D)

(1) S, S (2) S, R

(3) R, S (4) R, R

48. Zeise's salt has the formula $K[PtCl_3(C_2H_4)]\cdot H_2O$. How many different ligand types are coordinated to Pt in the coordination sphere?

(1) 3 (2) 1

(3) 4 (4) 2

49. What is the spin-only magnetic moment for a low-spin d^5 octahedral complex?

(1) 4.90 BM (2) 2.83 BM

(3) 1.73 BM (4) 5.92 BM

50. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Synthesis of ethyl phenyl ether may be achieved by Williamson synthesis.

Reason R: Reaction of bromobenzene with sodium ethoxide yields ethyl phenyl ether.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

51. Match **List-I** with **List-II**.

List-I		List-II	
(A)	–OH containing amino acid	(I)	Methionine
(B)	Acidic amino acid	(II)	Serine
(C)	Essential amino acid	(III)	Glycine
(D)	Optically inactive amino acid	(IV)	Glutamic acid

Choose the *most appropriate* answer from the options given below.

(1) A-I, B-IV, C-II, D-III

(2) A-II, B-IV, C-I, D-III

(3) A-II, B-IV, C-III, D-I

(4) A-IV, B-III, C-II, D-I

52. A mixture starts with equal initial concentrations of A and B, each decaying by first-order kinetics. The half-life of A is 300 s and that of B is 180 s. What is the value of the ratio $\frac{[A]_t}{[B]_t}$ after 900 s?

(Use ln 2 = 0.693.)

(1) 4 (2) $\frac{1}{2}$

(3) $\frac{1}{4}$ (4) 2

53. Given below are two statements.

Statement I: Complex compounds always produce ions when dissolved in solution.

Statement II: A complex ion remains intact in solution and does not dissociate into its constituent parts.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

54. Which statement must be **true** about the link between reaction order and mechanism?

(1) A zero order reaction cannot be a single step reaction.

(2) A first order reaction must be a single step reaction.

(3) A second order reaction cannot be a single step reaction.

(4) A zero order reaction can not be a complex reaction.

55. CO is a stronger ligand than Cl^- primarily because:
(1) CO is more reactive
(2) CO is a neutral molecule
(3) CO has π* orbitals
(4) CO is poisonous

56. Match **List-I** with **List-II**.

List-I (Orbitals)		List-II (Angular nodes)	
(A)	2s	(I)	3
(B)	5f	(II)	1
(C)	4d	(III)	2
(D)	3p	(IV)	0

Choose the *most appropriate* answer from the options given below.
(1) A-IV, B-I, C-II, D-III
(2) A-IV, B-I, C-III, D-II
(3) A-I, B-IV, C-III, D-II
(4) A-I, B-IV, C-II, D-III

57. Match **List-I** with **List-II**.

List-I (Equilibrium)		List-II (K_P/K_C)	
(A)	$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$	(I)	RT
(B)	$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$	(II)	$\frac{1}{(RT)^2}$
(C)	$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$	(III)	$\frac{1}{RT}$
(D)	$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$	(IV)	$(RT)^2$

Choose the *most appropriate* answer from the options given below.
(1) A-IV, B-III, C-II, D-I
(2) A-I, B-IV, C-III, D-II
(3) A-I, B-III, C-IV, D-II
(4) A-IV, B-III, C-I, D-II

58. When H_2 is ionized to H_2^+, identify from which molecular orbital the electron is removed and state the magnetic nature of H_2^+.
(1) From σ; diamagnetic
(2) From π (antibonding); paramagnetic
(3) From σ (bonding); paramagnetic
(4) From π (bonding); diamagnetic

59. An aqueous solution contains a solute with mole fraction $x_{solute} = 0.009$. What is the molality of the solution (in mol kg^{-1})?
(1) 0.40 m (2) 0.50 m
(3) 0.60 m (4) 0.30 m

60. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Boiling point of propan-1-ol is higher than the propane.
Reason R: In propan-1-ol, intermolecular hydrogen bonding exists.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

61. A 29% (w/w) aqueous H_2SO_4 solution has density = 1.96 g mL^{-1}. What is its molarity?
(1) 4.9 M (2) 5.8 M
(3) 5.0 M (4) 6.2 M

62. Atomic number of the element having symbol Uno is;
(1) 106
(2) 118
(3) 116
(4) 108

63. A 200 mg sample of CO_2 initially contains some number of molecules. After removing 1.204×10^{21} molecules, how many molecules of CO_2 are left?
(1) 1.53×10^{21}
(2) 8.60×10^{21}
(3) 1.33×10^{21}
(4) 9.80×10^{20}

64. An electron has a de Broglie wavelength λ. If its kinetic energy becomes nine times the original value, the new wavelength is:
(1) $\frac{\lambda}{9}$
(2) $\frac{\lambda}{3}$
(3) $\lambda\sqrt{9}$
(4) 3λ

[Contd...

65. Given below are two statements.
Statement I: The drop in first ionization enthalpy from B → Al is much larger than that from Al → Ga.
Statement II: In Gallium, the 3d subshell is completely filled.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

66. Which pair lists a property that increases down a group followed by one that decreases down a group (same order)?
(1) Atomic radius, electronegativity
(2) Electronegativity, atomic radius
(3) Electron gain enthalpy, atomic radius
(4) Electronegativity, electron gain enthalpy

67. Among $CO_3^{2-}, AsO_3^{3-}, XeO_3, ClO_3^-, BO_3^{3-}$ and SO_3^{2-} the non-planar species are:
(1) $AsO_3^{3-}, XeO_3, ClO_3^-, SO_3^{2-}$
(2) $BO_3^{3-}, CO_3^{2-}, SO_3^{2-}$
(3) $AsO_3^{3-}, BO_3^{3-}, CO_3^{2-}$
(4) $AsO_3^{3-}, XeO_3, CO_3^{2-}$

68. In the disproportionation of MnO_4^{2-} in acidic medium, manganese in the products appears in which pair of oxidation states?
(1) +7 and +4 (2) +6 and +2
(3) +4 and +2 (4) +7 and +2

69. Which among the following gases turn lime water milky?
(A) O_3
(B) SO_2
(C) NO_2
(1) B only
(2) A only
(3) B and C only
(4) A, B and C

70. For the salt AB_2, $K_{sp} = 6.91 \times 10^{-12}$. If at equilibrium $[A^{2+}] = 1.2 \times 10^{-4}$ M, the concentration of B^- is:
(1) 1.2×10^{-4} M (2) 2.4×10^{-4} M
(3) 6.91×10^{-6} M (4) 6.91×10^{-8} M

71. In the HCl addition to 3-chloro-1-butene, how many of the dichloro products are optically active?
(1) 2 (2) 0
(3) 1 (4) 3

72. Which sequence would convert cis-(Ph-CH=CH-CH_3) to the trans isomer?
(1) Br_2; alc KOH; $NaNH_2$; Na (liq NH_3)
(2) Br_2; alc KOH; $NaNH_2$; H_2, Lindlar catalyst
(3) H_2, Pt/C (excess)
(4) OsO_4, then $NaHSO_3/H_2O$

73. Hydration of but-1-yne in presence of $HgSO_4/H_2SO_4$ results the formation of:
(1) Butanal (2) Butan-2-one
(3) Butan-1-ol (4) Butan-2-ol

74. A non-volatile solute is dissolved in 5.0 mol of water. If the relative lowering of vapour pressure of the solution is $\frac{1}{4}$, how many moles of solute were added?
(1) 5/3 mol (2) 1.0 mol
(3) 2.0 mol (4) 3/2 mol

75. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: 2,2-Dimethylpropanal undergoes Cannizzaro reaction with conc. NaOH.
Reason R: Cannizzaro reaction is a disproportionation reaction.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

76. 60 g of a non-volatile solute of molar mass 120 g mol^{-1} is added to 180 g of water at constant T. What is the percentage lowering of vapour pressure of water?
(1) 4.76% (2) 5.00%
(3) 2.78% (4) 9.09%

77. At 298 K, a redox reaction transfers n = 2 electrons and has $E^\circ_{cell} = 0.177$ V. The value of K_c is:
(1) 10^6 (2) 10^3
(3) 10^4 (4) 10^2

[Contd...

78. In XeF_6, the central xenon atom has ___ lone pair(s) and exhibits ___ hybridization.
(1) 1; sp^3d^3
(2) 2; sp^3d^2
(3) 1; sp^3d^2
(4) 0; sp^3d^3

79. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The correct decreasing order of stability is: 2,3-dimethylbut-2-ene > 2-methylbut-2-ene > But-1-ene.
Reason R: More substituted alkene has the greater number of hyperconjugative C-H bonds and so greater is the stability of alkenes.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

80. How many unpaired electrons are present in Eu^{2+} and Ce^{3+}, respectively?
(1) 7 and 1
(2) 6 and 2
(3) 7 and 0
(4) 6 and 1

81. Match **List-I** with **List-II**.

List-I		List-II	
(A)	N,N-Dimethylethylamine	(I)	$CH_3-\underset{\displaystyle H}{\underset{\vert}{N}}-CH_2-CH_3$
(B)	Ethylmethylamine	(II)	$CH_3-CH_2-CH_2-NH_2$
(C)	Isopropylamine	(III)	$CH_3-\underset{\displaystyle NH_2}{\underset{\vert}{CH}}-CH_3$
(D)	n-Propylamine	(IV)	$CH_3-\underset{\displaystyle CH_3}{\underset{\vert}{N}}-CH_2-CH_3$

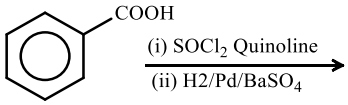
Choose the *most appropriate* answer from the options given below.
(1) A-IV; B-III; C-I; D-II
(2) A-I; B-IV; C-III; D-II
(3) A-IV; B-I; C-II; D-III
(4) A-IV; B-I; C-III; D-II

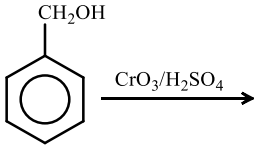
82. Given below are two statements:
Statement I: In solid state PCl_5 exist as an Ionic solid.
Statement II: The covalence of nitrogen in N_2O_5 is three.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

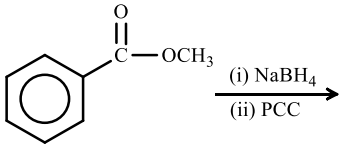
83. Benzene is first chlorinated with Cl_2/anh. $FeCl_3$, then the product is nitrated with conc. HNO_3/H_2SO_4. The final product is:
(1) 3-chloro-1-nitrobenzene
(2) 2-chloro-1-nitrobenzene
(3) 4-chloro-1-nitrobenzene
(4) A mixture of 2-chloro-1-nitrobenzene and 4-chloro-1-nitrobenzene

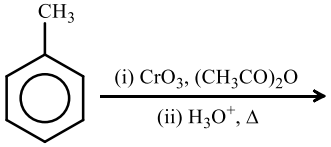
84. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Nitrogen present in nitrobenzene cannot be determined by Kjeldahl method.
Reason R: Nitrogen present in nitrobenzene does not change to ammonium sulphate.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

85. Which of the following reactions will yield benzaldehyde as a product?

(A) 

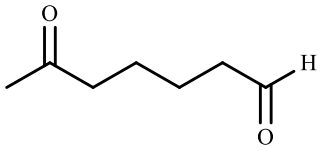
(B) 

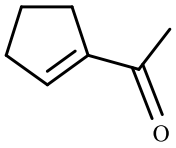
(C) 

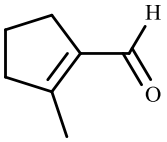
(D) 

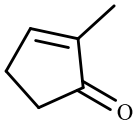
(1) (B) and (C) (2) (C) and (D)
(3) (A) and (D) (4) (A) and (C)

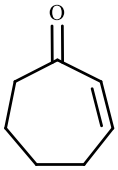
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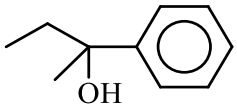
86. When  undergoes intramolecular aldol condensation, the major product formed is:

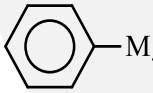
(1) 

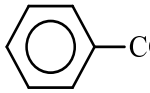
(2) 

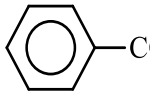
(3) 

(4) 

87.  can be prepared by:

A. $CH_3CH_2COCH_3$ +  followed by hydrolysis

B.  + CH_3MgBr followed by hydrolysis

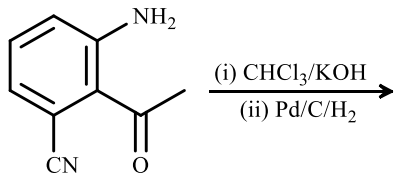
C.  + C_2H_5MgBr followed by hydrolysis

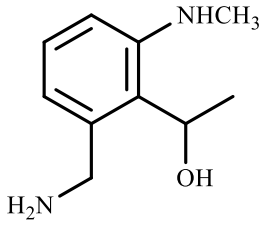
(1) A and B only
(2) B and C only
(3) A and C only
(4) A, B and C

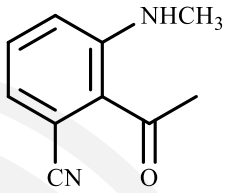
88. Which pair consists of vitamins that are not stored for long and are readily excreted (i.e., water-soluble)?

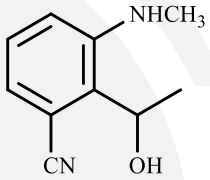
(1) Thiamine (B_1) and Ascorbic acid (Vitamin C)
(2) Vitamin A and Vitamin D
(3) Vitamin D and Vitamin K
(4) Vitamin A and Vitamin E

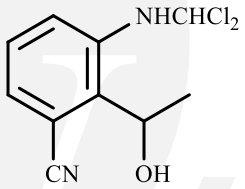
89. The major product obtained in the following reaction is:



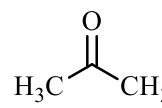
(1) 

(2) 

(3) 

(4) 

90. Match **List-I** with **List-II**.

	List-I (Chemical Reaction)		**List-II (Reagent used)**
(A)	$CH_3COOCH_2CH_3 \rightarrow CH_3CH_2OH$	(I)	CH_3MgBr/H_3O^+ (1 equivalent)
(B)	$CH_3COOCH_3 \rightarrow CH_3CHO$	(II)	H_2SO_4/H_2O
(C)	$CH_3C \equiv N \rightarrow CH_3CHO$	(III)	DiBAL-H/H_2O
(D)	$CH3C \equiv N \rightarrow$ 	(IV)	$SnCl_2$, HCl/H_2O

Choose the *most appropriate* answer from the options given below.

(1) A-II, B-IV, C-III, D-I
(2) A-II, B-III, C-IV, D-I
(3) A-III, B-II, C-I, D-IV
(4) A-IV, B-II, C-III, D-I

[Contd...

91. Match **List-I** with **List-II** regarding nomenclature of *Eco*RI

List-I		List-II	
(A)	E	(I)	Indicate the order in which the enzymes were isolated from that strain of bacteria
(B)	co	(II)	Name of strain
(C)	R	(III)	Represent the species of source bacteria
(D)	I	(IV)	Represent the genus of source bacteria

Choose the **correct** answer from the options given below:

(1) A-IV, B-III, C-I, D-II

(2) A-IV, B-III, C-II, D-I

(3) A-I, B-II, C-III, D-IV

(4) A-IV, B-II, C-I, D-III

92. Due to a mutation, a yeast cell produces defective RNA polymerase III. What will be its most likely consequence?

(1) Synthesis of 5.8S rRNA in it will stop.

(2) The catalytic RNA will not be produced inside its mitochondria.

(3) Polypeptide synthesis in cytoplasm will not take place.

(4) Polynucleotide synthesis in nucleus will be compromised.

93. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Number of male gametes in a pollen grain	(I)	7
(B)	Number of cells in an embryo sac	(II)	2
(C)	Number of nuclei in an embryo sac	(III)	8
(D)	Number of egg cells in an embryo sac	(IV)	1

Choose the ***most appropriate*** answer from the options given below.

(1) A-II, B-III, C-I, D-IV

(2) A-II, B-I, C-III, D-IV

(3) A-III, B-I, C-II, D-IV

(4) A-IV, B-I, C-III, D-II

94. Read the following characteristics about basidiomycetes.

A. Asexual spores are generally formed.

B. Vegetative reproduction by fragmentation is common.

C. The sex organs are absent, but plasmogamy is brought about by fusion of two vegetative or somatic cells of different strains.

D. Karyogamy and meiosis take place in basidium to form four basidiospores.

E. The basidiospores are endogenously produced in the basidium.

Identify the option with the **correct** statements.

(1) A, B and C

(2) B, C and D

(3) C, D and E

(4) A, D and E

95. Which of the following statements are **correct**?

A. Invertebrates possess very simple endocrine systems with few hormones.

B. The hypothalamus is the upper part of the forebrain.

C. The hypothalamus contains several groups of neurosecretory cells which produce hormones.

D. The hypothalamus produces releasing hormone, inhibiting hormone, oxytocin and vasopressin.

E. Somatostatin from the hypothalamus stimulates the release of growth hormone from the pituitary.

(1) A, B and C only

(2) A, C and D only

(3) A, B, C and D

(4) B, C, D and E

96. Which is **not** true related to *Bt* genes?

(1) It acts as bio-pesticide.

(2) It makes plants susceptible to pests.

(3) It decreases the amount of pesticide used in plants.

(4) It is obtained from bacteria.

97. Which of the following is a function of the mesosome in a prokaryotic cell?

(1) Synthesis of proteins

(2) Trapping light energy

(3) DNA replication and distribution to daughter cells

(4) Motility

98. A researcher is studying the decomposition rates of two different types of detritus in two different experimental setups:
Detritus A: Rich in lignin and chitin.
Detritus B: Rich in nitrogen and water-soluble sugars.
Setup I: The detritus is kept in a warm, moist environment with high aeration.
Setup II: The detritus is kept in a low-temperature environment with anaerobic conditions.
Based on the principles of decomposition, which of the following statements is most likely to be **correct**?
(1) The rate of decomposition for Detritus A in Setup I will be faster than Detritus B in Setup II, because environmental factors always override chemical composition.
(2) Detritus B in Setup I will show the highest rate of mineralization because both the chemical quality of the detritus and the climatic conditions are synergistic for microbial activity.
(3) Humification will occur faster in Setup II for Detritus B than in Setup I for Detritus A, because low temperatures promote the accumulation of dark amorphous substances.
(4) In Setup II, the rate of catabolism will be higher than the rate of fragmentation because bacterial enzymes are less affected by oxygen availability than detritivores like earthworms.

99. Read the following statements related to *Bt* spores used as biocontrol agent.
A. Mixed with organic solvent.
B. They are available in sachets as dried spores
C. Sprayed onto vulnerable plants such as brassicas and fruit trees.
D. In the gut of the larvae, the toxin is released and the larvae get killed.
Identify the **correct** option.
(1) Only A and C are correct
(2) Only A is incorrect
(3) Only B and D are incorrect
(4) A, B and C are correct

100. Which of the following characteristics are **not** associated with wind-pollinated flowers?
A. Light and non-sticky pollen grains
B. Large, colourful flowers
C. Well-exposed stamens
D. Production of nectar
E. Large, often-feathery stigma
(1) A and C only (2) B and D only
(3) A, C and E only (4) B, D and E only

101. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Origin of universe	(I)	4.5 bya
(B)	Origin of earth	(II)	13.8 bya
(C)	Origin of life on earth	(III)	2000 mya
(D)	First cellular form of life	(IV)	4 bya

Choose the **correct** answer from the options given below:
(1) A-II, B-I, C-III, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-III, B-I, C-II, D-IV
(4) A-III, B-I, C-IV, D-II

102. Intermediate state structure of the substrate, formed during an enzymatic reaction is:
(1) transient and unstable.
(2) permanent and stable.
(3) transient and stable.
(4) permanent and unstable.

103. Read the following statements about Gel Electrophoresis and choose the **correct** statement.
A. Since DNA is negatively charged, it will move towards a positively charged cathode.
B. Agarose is a synthetic polymer that is extracted from seaweeds.
C. Large-sized DNA fragments move much nearer to the anode.
D. On staining with ethidium bromide and under UV rays, DNA gives orange colour bands.
(1) A, B, C and D (2) B and D only
(3) A, B and D only (4) D only

104. Which of the following characters are expressed in both homozygous and heterozygous conditions?
A. Axial position of flower in pea plant
B. Violet flower colour in pea plant
C. Pink colour flower in snapdragon
D. Round seed shape in pea plant
E. Green seed colour in pea plant
Choose the **correct** answer from the options given below:
(1) A, B and E only
(2) A, B and D only
(3) B, C and D only
(4) A, C and D only

105. In a normal ECG, which of the following represents the depolarisation of the ventricles?
(1) P-wave (2) QRS complex
(3) T-wave (4) End of T-wave

[Contd...

106. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Amniocentesis	(I)	Assist those couple which are childless
(B)	Lactational amenorrhea	(II)	Induced abortion
(C)	MTP	(III)	Detection of genetic disorder in unborn foetus
(D)	Assisted reproductive technologies	(IV)	Absence of menstruation due to intense lactation

Choose the **correct** answer from the options given below:

(1) A-IV, B-II, C-III, D-I
(2) A-III, B-IV, C-II, D-I
(3) A-II, B-IV, C-I, D-III
(4) A-II, B-IV, C-III, D-I

107. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Fighting fish	(I)	*Pterophyllum*
(B)	Angel fish	(II)	*Betta*
(C)	Flying fish	(III)	*Hippocampus*
(D)	Sea horse	(IV)	*Exocoetus*

Choose the **correct** answer from the options given below:

(1) A-II, B-I, C-IV, D-III
(2) A-II, B-III, C-IV, D-I
(3) A-IV, B-I, C-II, D-III
(4) A-II, B-IV, C-I, D-III

108. Match **List-I** with **List-II**.

List-I (Hormone)		List-II (Function)	
(A)	Parathyroid hormone (PTH)	(I)	Regulates water and electrolyte balance
(B)	Aldosterone	(II)	Produces anti-inflammatory reactions and suppresses immune response
(C)	Adrenal Androgens	(III)	Increase blood calcium levels
(D)	Cortisol	(IV)	Growth of axial, pubic, and facial hair during puberty

Choose the **correct** answer from the options given below:

(1) A-III, B-I, C-IV, D-II
(2) A-II, B-I, C-IV, D-III
(3) A-III, B-II, C-I, D-IV
(4) A-II, B-III, C-I, D-IV

109. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Squamous epithelium	(I)	Protection against chemical and mechanical stresses
(B)	Cuboidal epithelium	(II)	Movement of particle in specific direction
(C)	Compound epithelium	(III)	Formation of diffusion boundary
(D)	Ciliated epithelium	(IV)	Secretion and absorption

Choose the **correct** answer from the options given below:

(1) A-I, B-IV, C-II, D-III
(2) A-III, B-IV, C-I, D-II
(3) A-II, B-I, C-III, D-IV
(4) A-IV, B-I, C-III, D-II

110. Which of the following pairs of animals has non glandular skin?

(1) Snake and Frog
(2) *Chameleon* and Turtle
(3) Frog and Pigeon
(4) Crocodile and Tiger

111. Which of the following functions are under the control of centres located in medulla oblongata?

A. Cardiovascular reflexes
B. Respiration
C. Gastric secretions
D. Urge for drinking

(1) A and B only
(2) B and C only
(3) A, B, C and D
(4) A, B and C only

112. Match **List-I** with **List-II** w.r.t. number of different vertebrae in vertebral column.

List-I		List-II	
(A)	Cervical	(I)	12 in number
(B)	Thoracic	(II)	1 in number after fusion
(C)	Coccygeal	(III)	7 in number
(D)	Lumbar	(IV)	5 in number

Choose the **correct** answer from the options given below:

(1) A-III, B-II, C-IV, D-I
(2) A-I, B-IV, C-II, D-III
(3) A-IV, B-I, C-II, D-III
(4) A-III, B-I, C-II, D-IV

[Contd...

113. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Heroin	(I)	Pain killer
(B)	*Datura*	(II)	Coke
(C)	Cocaine	(III)	Hallucinogen
(D)	Morphine	(IV)	Depressant

Choose the **correct** answer from the options given below:

(1) A-II, B-I, C-III, D-IV
(2) A-III, B-II, C-IV, D-I
(3) A-IV, B-III, C-II, D-I
(4) A-I, B-II, C-IV, D-III

114. Which of the following statements is **correct** for atherosclerosis?

(1) Lumen of the vein that supplies blood to the heart muscle gets wider.
(2) Obstruction in the blood vessel leads to an increase in blood flow to heart muscle.
(3) It is caused by the deposition of fats, and cholesterol in walls of the vein.
(4) Build up of calcium, fat and fibrous tissue makes the lumen of coronary arteries narrower.

115. In honey bees, males:

(1) have 32 chromosomes.
(2) are diploid.
(3) are produced by parthenogenesis.
(4) have same chromosome number like worker bees.

116. In the overall process of glycolysis, which of the following statements accurately describes the net yield for the complete conversion of one molecule of glucose?

(1) 2 molecules of Pyruvic acid, net gain of 2 ATP, and 2 NADH + H^+
(2) 1 molecule of Pyruvic acid, net gain of 4 ATP, and 2 NADH + H^+
(3) 2 molecules of Pyruvic acid, net gain of 2 ATP, and 1 NADH + H^+
(4) 2 molecules of Pyruvic acid, net gain of 2 ATP, and 4 NADH + H^+

117. A colour-blind female marries a man whose father is normal but mother was colour-blind, what is the probability that their children (both sons and daughters) will be colour-blind?

(1) 25% only (2) 50% only
(3) 75% only (4) 100%

118. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The Indian Parliament has recently cleared the first amendment of the Indian Patents Bill.
Reason R: The Indian Patent Bill takes many issues into consideration, including patent terms, emergency provisions and research and development initiatives.
In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is NOT the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

119. The descriptions below outline the nutritional characteristics of members in each of the 5 kingdoms.

A. Exclusively heterotrophic (saprophytic /parasitic) with absorptive nutrition.
B. Mostly autotrophic (photosynthetic), except a few.
C. Both autotrophic (photosynthetic) and heterotrophic members exist.
D. Exclusively heterotrophic (holozoic /saprophytic) with ingestive nutrition.
E. Show the most extensive metabolic diversity, including photosynthetic, chemosynthetic, and heterotrophic modes.

Arrange the above characteristics in an order that represents the mode of nutrition for Monera, Protista, Fungi, Plantae, and Animalia respectively, and choose the **correct** option.

(1) E, C, A, B, D
(2) E, C, B, A, D
(3) C, E, A, B, D
(4) E, A, C, B, D

120. A student observes a transverse section of a plant specimen and notes the following features: conjoint and closed vascular bundles, absence of phloem parenchyma, and presence of water-containing cavities. The specimen is most likely a:

(1) dicot root.
(2) monocot stem.
(3) dicot stem.
(4) monocot leaf.

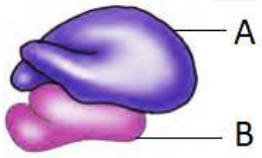
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121. Read the following statements regarding thyroid gland and choose **incorrect** ones.

A. Its two lobes are interconnected with a thin flap of connective tissue called isthmus.
B. It is composed of follicles and stromal tissues.
C. Iodine is essential for the normal rate of hormone synthesis in the thyroid.
D. Due to cancer of the thyroid gland, the rate of synthesis and secretion of the thyroid hormones decreases to abnormal low levels.

(1) A, B and C
(2) B and C only
(3) B, C and D
(4) D only

122. Go through the given diagram of a cell organelle and choose the incorrect statement about it.



(1) This organelle is present in all cells.
(2) It is composed of a nucleic acid and proteins and surrounded by a single membrane.
(3) In human cell cytoplasm, 'A' is 60S.
(4) In a bacterial cell, 'B' is 30S.

123. If the sequence of coding strand of DNA is 5'GCATGCATGCATT3', then what will be the sequence of its primary transcript?

(1) 3' CGUAGCUACGUAU 5'
(2) 3' GCAUGCAUGCAUU 5'
(3) 5' CGUACGUACGUAA 3'
(4) 5' GCAUGCAUGCAUU 3'

124. Which of the following features is **not** related to pons?

(1) It is a part of the brain stem.
(2) It consists of fibre tracts that interconnect different regions of the brain.
(3) It is found in the hind brain.
(4) It has a very convoluted surface in order to provide the additional space for many more neurons.

125. Respiratory part of the respiratory system in human is formed by;

(1) external nostrils up to the alveoli.
(2) external nostrils up to initial bronchioles.
(3) trachea to terminal bronchioles.
(4) alveoli and its ducts.

126. Given below are two statements:

Statement I: Evolutionary biology is the study of history of life forms on earth.

Statement II: The Big Bang theory attempts to explain to us the origin of the universe.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct, but Statement II is incorrect.
(2) Statement I and Statement II both are correct.
(3) Statement I is incorrect, but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

127. Match **List-I** with **List-II**.

List-I (Organism)		**List-II (Characteristic)**	
(A)	*Chara*	(I)	Oogamous sexual reproduction, chloroplasts contain *a* and *d*
(B)	*Anabaena*	(II)	Stored food is laminarin or mannitol, chloroplasts contain *a* and *c*
(C)	*Polysiphonia*	(III)	Chloroplasts contain pyrenoids, chlorophyll *a* and *b*
(D)	*Dictyota*	(IV)	Chloroplasts are not present, can fix atmospheric nitrogen in specialised cells

Choose the ***most appropriate*** answer from the options given below.

(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-I, B-II, C-III, D-IV
(4) A-II, B-I, C-IV, D-III

128. Given below are two statements:

Statement I: Malvaceae is commonly called the cotton family.

Statement II: The fruit in members of family Graminae is generally a capsule.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

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129. Identify the **incorrect** statement.
(1) Spraying juvenile conifers with GAs hastens the maturity period.
(2) Auxin controls xylem differentiation.
(3) ABA helps seeds to withstand desiccation and other factors unfavourable for growth.
(4) Cytokinins help overcome the lateral dominance.

130. What is common between C_3 and C_4 plants?
(1) Primary CO_2 acceptor
(2) Sugar synthesis by Calvin cycle
(3) Presence of Kranz anatomy in leaves
(4) Amount of biomass produced under same conditions

131. Which of the following parts passes sperms from Bidder's canal to cloaca in male frog?
(1) Vasa efferentia (2) Urinogenital duct
(3) Cloacal aperture (4) Testis

132. In the nephrons, glucose is reabsorbed in the blood at:
(1) Glomerulus, by selective reabsorption.
(2) Collecting duct, by active secretion.
(3) Proximal convoluted tubule, by active transport.
(4) Distal convoluted tubule, by passive diffusion.

133. Primary steps that can be taken towards attaining reproductive health are;
A. Counselling and creating awareness among people about reproductive organs, adolescence and associated changes.
B. Safe and hygienic sexual practices.
C. Creating awareness about STDs including AIDS.
D. Ban on MTP and amniocentesis.
(1) A and B only (2) B and C only
(3) A, B and C only (4) A, B, C and D

134. Read the following statements (A-D).
A. *In-situ* conservation includes National Parks and Wildlife Sanctuaries.
B. Sacred groves are found in Khasi and Jaintia Hills of Meghalaya.
C. Cryopreservation is an example of *in-situ conservation*.
D. The Earth Summit (1992) was held in Rio de Janeiro.
How many of the above statements are **correct**?
(1) Four (2) One only
(3) Two only (4) Three only

135. Given below are two statements:
Statement I: Ecology is one of the threads which gives us a holistic perspective to biology.
Statement II: A particular aspect of ecology is the study of anthropogenic environmental degradation and the socio-political issues it has raised.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is **incorrect** but Statement II is **correct**.

136. Few symptoms are given below:
Choose the **correct** disease in which all these symptoms are present.
A. Internal bleeding
B. Muscular pain
C. Fever
D. Anemia
E. Blockage of intestinal passage
(1) Typhoid (2) Ascariasis
(3) Malaria (4) Pneumonia

137. Given below are two statements:
Statement I: Pyriform gametes with two apical flagella are present in members of phaeophyceae.
Statement II: Post-fertilisation changes do not occur in members of rhodophyceae.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

138. If in an initial population of 375 individuals, 40 died, 56 went to other habitats, 72 were born and 32 came from elsewhere in a period of 1 year; the net change in population;
(1) is 0. (2) can't be determined.
(3) is increase by 8. (4) is decrease by 8.

139. The presence of a tetradynamous androecium is characteristic features of which family?
(1) Solanaceae (2) Malvaceae
(3) Gramineae (4) Brassicaceae

140. Which of the following represents the **correct** complementary (antisense) strand for the *Eco*RI recognition sequence?
(1) 5'-GAATTC-3' (2) 3'-CCCGGG-5'
(3) 5'-GGATCC-3' (4) 3'-CTTAAG-5'

141. Match **List-I** with **List-II** w.r.t cross and their phenotypic ratio.

List-I		List-II	
(A)	RrYy × rryy	(I)	9 : 3 : 3 : 1
(B)	RrYy × RrYy	(II)	1: 1 : 1: 1
(C)	Rr × Rr	(III)	1: 1
(D)	Rr × rr	(IV)	3: 1

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-I, D-IV
(2) A-III, B-I, C-IV, D-II
(3) A-II, B-I, C-IV, D-III
(4) A-III, B-IV, C-II, D-I

142. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Parturition	(I)	Attachment of blastocyst to endometrium
(B)	Gestation	(II)	Release of egg
(C)	Ovulation	(III)	Delivery of baby
(D)	Implantation	(IV)	Period between fertilisation and birth

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-I, D-II
(3) A-II, B-I, C-IV, D-III
(4) A-III, B-IV, C-II, D-I

143. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Cytoplasmic increase in a cell occurs intermittently during cell division.
Reason (R): The events of cell growth and its division are under genetic control.
In the light of the above statements, choose the ***most appropriate*** from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

144. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Casparian strips	(I)	Large, empty, colourless cells in the adaxial epidermis of grasses
(B)	Stomata	(II)	Unicellular elongations of epidermal cells
(C)	Bulliform cells	(III)	Root endodermis walls deposited with water-impermeable, waxy material suberin
(D)	Root hairs	(IV)	Regulate the process of transpiration and gaseous exchange

Choose the ***most appropriate*** answer from the options given below.
(1) A-II, B-IV, C-I, D-III
(2) A-III, B-II, C-I, D-IV
(3) A-II, B-I, C-III, D-IV
(4) A-III, B-IV, C-I, D-II

145. Which of the following is **incorrect** about ABO blood group system?
(1) The alleles I^A and I^B produce a slightly different form of the sugar, while allele *i* does not produce any sugar.
(2) There are 6 phenotypes and 4 genotypes possible from the combination of alleles.
(3) It shows multiple allelism, but all three alleles cannot be seen in a single individual.
(4) The plasma membrane of the red blood cells has sugar polymers that protrude from its surface and the kind of sugar is controlled by the gene *I*.

146. Which of the following statements is/are **correct**?
A. Reduced levels of progesterone resulted in osteoporosis in old people.
B. Number of ribs in humans is 24.
C. Muscular dystrophy occurs due to the accumulation of uric acid crystals.
D. Joint between atlas and axis is an example of synovial joint.
E. Pectoral and Pelvic girdle bones help in the articulation of the upper and the lower limbs respectively with the axial skeleton.
(1) A, B and C only
(2) B and D only
(3) A, C, D and E
(4) B, D and E only

[Contd...

147. Which of the following have the same number of chromosomes as that of spermatid?

A. Secondary oocyte
B. Secondary spermatocyte
C. Ootid
D. Ovum

(1) C and D only (2) A and D only
(3) D only (4) A, B, C and D

148. There are three basic steps in genetically modifying an organism. Arrange these steps in the **correct** sequence.

A. Introduction of the identified DNA into the host.
B. Maintenance of introduced DNA in the host and transfer of the DNA to its progeny.
C. Identification of DNA with desirable genes.

(1) A → B → C
(2) B → C → A
(3) C → B → A
(4) C → A → B

149. Identify the **correct** statement.

(1) Theodore Schwann, a botanist, concluded that the presence of cell wall is a unique character of the plant cells.
(2) Matthias Schleiden, a zoologist, observed that all plants are composed of different kinds of cells which form the tissues of the plant.
(3) Antonie Von Leeuwenhoek first saw and described a live cell using an electron microscope.
(4) Rudolf Virchow modified the hypothesis of Schleiden and Schwann to give the cell theory a final shape by explaining that cells arise from pre-existing cells.

150. Select the **incorrect** option w.r.t. disorder of the circulatory system in humans.

(1) High blood pressure affects vital organs like the brain and kidney.
(2) Heart failure means the state of heart when it is not pumping blood effectively enough to meet the needs of the body.
(3) Angina is more common among the middle-aged and elderly.
(4) When the heart muscle is suddenly damaged by an inadequate blood supply, it is referred to as cardiac arrest.

151. Which of the following statements regarding Whittaker's five-kingdom classification is **correct**?

(1) The main criteria of classification included morphological characters but did not consider phylogenetic relationship amongst organisms.
(2) It grouped together prokaryotic bacteria with eukaryotic fungi.
(3) Unicellular and multicellular organisms like *Chlamydomonas* and *Spirogyra* were placed in the same group as algae.
(4) It placed fungi in a separate kingdom, Kingdom Fungi, based on their chitinous cell wall and heterotrophic nature.

152. Which of the following mechanisms can be studied in sliding filament theory?

(1) Shortening of 'I' band while 'A' band remains unaffected.
(2) Shortening of actin and myosin filament while sliding over each other.
(3) Shortening of H zone while I band remains intact.
(4) Shortening of myosin filament while actin filament remains unchanged.

153. Select the **correct** hierarchical sequence for *Homo sapiens* from species to kingdom.

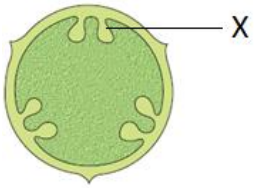
(1) *sapiens* → *Homo* → Hominidae → Primata → Mammalia → Chordata → Animalia
(2) *Homo* → *sapiens* → Mammalia → Primata → Animalia → Chordata → Hominidae
(3) *Homo* → *sapiens* → Primata → Mammalia → Hominidae Chordata → Animalia
(4) *sapiens* → *Homo* → Hominidae → Mammalia → Animalia → Primata → Chordata

154. Choose the **incorrectly** matched pair.

(1) Wheat and mango: Same class but different division
(2) *nigrum* and *melongena*: Same genus and same kingdom
(3) *Petunia* and *Datura:* Different genera but same family
(4) Man and housefly: Same kingdom but different phyla

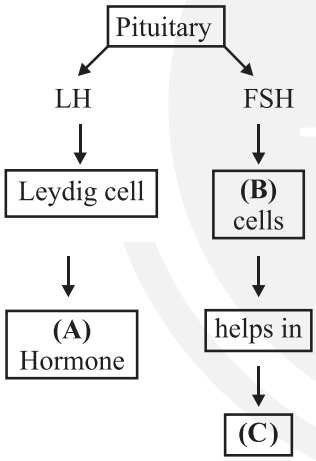
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155. Shown below is a type of placentation. Choose the correct statement about it.



(1) The ovary here is one-chambered but becomes two-chambered due to the formation of false placenta.
(2) The label 'X' refers to the structure that becomes seed after fertilisation.
(3) Such type of placentation is seen in tomato and lemon.
(4) In this type of placentation, the placenta develops at the base of ovary.

156. Given below is an incomplete flowchart showing the influence of hormones on gametogenesis in males. Observe the flowchart carefully and identify **A**, **B** and **C**.



	A	B	C
(1)	Progesterone	Follicular	Spermatogenesis
(2)	GnRH	Follicular	Spermiogenesis
(3)	GnRH	Sertoli	Spermatogenesis
(4)	Androgens	Sertoli	Spermiogenesis

157. Identify the **correct** order of abundance of chemical constituents in living cells.
(1) Water > Carbohydrates >Lipids > Proteins > Nucleic acids > Ions
(2) Water > Nucleic acids >Lipids > Proteins > Carbohydrates > Ions
(3) Water > Proteins > Nucleic acids > Carbohydrates > Lipids > Ions
(4) Proteins > Water > Carbohydrates > Lipids > Ions > Nucleic acids.

158. Given diagram shows a cotton plant with pest infestataion, this can be controlled by the proteins encoded by the gene:



(1) *cryIAc* only
(2) *cryIIAb* only
(3) *cryIAb* and *cryIIAb*
(4) *cryIAc* and *cryIIAb*

159. Which of the following statement is/are **correct** regarding nerve impulse conduction?
A. The inner membrane potential changes from positive to negative, during depolarisation.
B. The electrical potential difference across the depolarised plasma membrane is called the resting potential.
C. In a resting neuron, the axoplasm inside the axon contains high concentration of Na^+.
D. Rise in stimulus-induced permeability to Na^+ is responsible for depolarisation.
(1) A and B only
(2) B only
(3) C and D only
(4) D only

160. Given below are two statements:

Statement I: Adults of *Periplaneta americana* are about 34-53 mm long.

Statement II: Only in the first two segments of cockroach, exoskeleton has hardened plates called sclerites.

In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct, but Statement II is incorrect.
(2) Statement I and Statement II both are correct.
(3) Statement I is incorrect, but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

[Contd...

161. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Red algae	(I)	Classified as true bacteria
(B)	Brown algae	(II)	Stored food is starch
(C)	Green algae	(III)	Spores or gametes are non-motile
(D)	Blue-green algae	(IV)	Union of gametes may take place in water or inside oogonium

Choose the ***most appropriate*** answer from the options given below.
(1) A-III, B-IV, C-II, D-I
(2) A-IV, B-II, C-III, D-I
(3) A-I, B-III, C-II, D-IV
(4) A-III, B-I, C-IV, D-II

162. Today, transgenic models exist for many human diseases which include
A. Cancer
B. Cystic fibrosis
C. Rheumatoid arthritis
D. Alzheimer's disease
(1) A and C only (2) B and C only
(3) A, B and C only (4) A, B, C and D

163. Which of the following is an ideal contraceptive for the female who wants to delay pregnancy and/or space children?
(1) Barrier method
(2) Intra Uterine Devices
(3) Lactational amenorrhea
(4) Surgical method

164. Amongst the following places, the highest number of bird species is found in;
(1) Colombia. (2) India.
(3) New York. (4) Greenland.

165. Read the following statements regarding the *lac* operon.
A. Lactose is the substrate for the enzyme beta-galactosidase and it regulates switching on and off of the operon.
B. The repressor of the operon is synthesised constitutively from the *z* gene.
C. In the presence of the inducer, the repressor is activated by interaction with it.
D. Regulation of *lac* operon by repressor is referred to as negative regulation.
E. The *y* gene codes for transacetylase.
In the light of above statements choose the **correct** answer from the options given below:
(1) B and E are correct, A, C and D are incorrect
(2) C and D are correct, A, B and E are incorrect
(3) A and D are correct, B, C and E are incorrect
(4) A, C and E are correct, B and D are incorrect

166. Choose the **correct** statement from the options given below.
(1) Branchial respiration occurs in sponges.
(2) Insect uses cuticle for exchange of gases.
(3) Workers in grinding and stone breaking industries may suffer from lung fibrosis.
(4) We have one external nostril.

167. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Pigments	(I)	Lemon grass oil
(B)	Alkaloids	(II)	Monoterpenes
(C)	Terpenoids	(III)	Morphine
(D)	Essential oil	(IV)	Carotenoids

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-I, D-IV
(2) A-I, B-II, C-III, D-IV
(3) A-II, B-I, C-III, D-IV
(4) A-IV, B-III, C-II, D-I

168. A diploid female plant (2n) is crossed with a tetraploid male plant (4n). The ploidy of the endosperm will be:
(1) 3n (2) 4n
(3) 5n (4) 2n

169. Match **List-I** with **List-II** regarding frog.

List-I		List-II	
(A)	Bidder's canal	(I)	Secretes bile
(B)	Mesorchium	(II)	Connects testes to kidney
(C)	Cloaca	(III)	Present inside the kidney
(D)	Liver	(IV)	Common chamber for faeces and urine

Choose the **correct** answer from the options given below:
(1) A-III, B-II, C-I, D-IV
(2) A-III, B-II, C-IV, D-I
(3) A-II, B-III, C-I, D-IV
(4) A-II, B-III, C-IV, D-I

170. Given below are two statements:
Statement I: DNA is a hydrophilic molecule and thus cannot pass through cell membrane easily.
Statement II: Micro-injection is a technique which introduces rDNA into the nucleus of plant cells.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

171. Consider the following statements regarding population interactions and select the option with all **correct** statements.
A. In predation, only one species benefits and the interaction is detrimental to the other.
B. In commensalism, one species is benefitted and the other is neither benefitted nor harmed.
C. In amensalism, both species are harmed.
D. In competition, both the interacting species lose.
E. In mutualism, one species is harmed while the other is unaffected.
(1) B, C and D (2) A, C and E
(3) A, B and D (4) A, D and E

172. Mark the **correct** statements.
A. Yoga has been practised to achieve physical and mental health.
B. Infectious diseases are very common and everyone of us suffers from these at sometime or other.
C. AIDS is an infectious disease.
D. Saliva in the mouth is an example of a physiological barrier of innate immunity.
E. Healthy persons bring economic prosperity.
(1) A, B and C only (2) B, C, D and E only
(3) C and D only (4) A, B, C, D and E

173. During the light reaction, FNR is;
(1) active during non-cyclic photophosphorylation only.
(2) the enzyme that catalyses reduction of NADPH + H^+.
(3) present in the stromal lamellae.
(4) the primary electron acceptor from PS II.

174. For photosynthesis, there is a _____ relationship between incident light and CO_2 fixation rates at low light intensities.
(1) inverse (2) linear
(3) hyperbolic (4) parabolic

175. Two leaves A and B have surface area 50 cm^2 and 100 cm^2, respectively. They both increase their surface area by 5 cm^2 in a day. The absolute growth rate;
(1) of A is more than B.
(2) of B is more than A.
(3) is 5 cm^2 per day.
(4) is 0.1 and 0.05 for leaf A and B, respectively.

176. After telophase I during meiosis;
(1) chromosomes do not undergo dispersion.
(2) chromosomes reach the extremely extended state of interphase nucleus.
(3) the cell exits the cell cycle to enter a quiescent stage.
(4) the cell undergoes cytokinesis to form a dyad of haploid cells.

177. The complex of mitochondrial ETS that does **not** have Fe-S centre is;
(1) complex I. (2) complex II.
(3) complex III. (4) complex IV.

178. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): The main function of phloem is to transport photosynthates.
Reason (R): Phloem transports food materials usually from roots to the leaves of plants.
In the light of the above statements, choose the most appropriate answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

179. Match **List-I** with **List-II**.

List-I		**List-II**	
(A)	*Oscillatoria*	(I)	Yeast used in making bread and beer
(B)	*Nucleopolyhedrovirus*	(II)	Biofertiliser
(C)	*Saccharomyces cerevisiae*	(III)	Yeast, source of statins
(D)	*Monascus purpureus*	(IV)	Bicontrol agent

Choose the ***most appropriate*** answer from the options given below.
(1) A-II, B-IV, C-I, D-III
(2) A-IV, B-II, C-III, D-I
(3) A-II, B-I, C-IV, D-III
(4) A-III, B-IV, C-I, D-II

180. Match **List-I** with **List-II**.

List-I		**List-II**	
(A)	Rheumatoid arthritis	(I)	Antitoxin
(B)	Tetanus	(II)	Autoimmune disorder
(C)	Carcinogen	(III)	Neoplastic transformation
(D)	Malignant tumor	(IV)	Metastasis

Choose the **correct** answer from the options given below:
(1) A-II, B-I, C-III, D-IV
(2) A-III, B-II, C-IV, D-I
(3) A-IV, B-III, C-II, D-I
(4) A-I, B-II, C-IV, D-III

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